

HW 1.2c

Name(s): _____

1. Suppose that Awesomeness (A) is related to Coolness (C) and Niceness (N) using the following:

$$\boxed{A^2 = 5C^3N^{1.5}}$$

A. Write a proportionality relation that relates Awesomeness and Niceness

B. Rewrite the equation in relative form for Awesomeness

C. Nice triples, and Cool stays the same, Awesomeness will _____

D. Rewrite the equation in relative form for Cool

2. Newton's Law of Gravity is $\boxed{F = \frac{GMm}{r^2}}$, where $G = 6.67 \times 10^{-11}$ is a constant, M and m are the masses of two objects, and r is the distance they are separated.

A. Write a proportionality relation that relates F and M

B. If M triples, F will _____.

C. Write a proportionality relation that relates F and r

D. If r triples, F will _____.

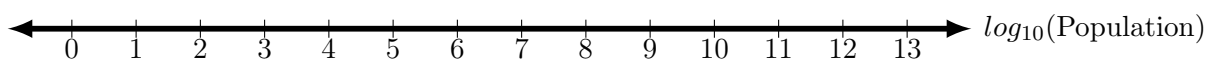
E. Rewrite Newton's Law of Gravity in **Relative Form**

F. If M triples, m is halved, and r quadruples, F will _____

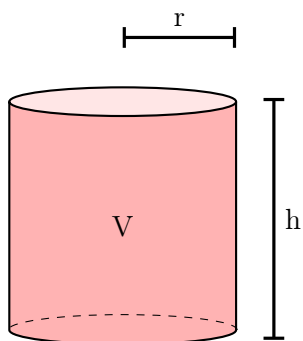
3. Populations can range over many orders of magnitude when we aggregate over different geographic extents.

Category	Estimate
1. Average occupants of one house in the USA	2.5 people per household
2. Average occupants per block in Portland, OR	80 people per block
3. Population of Portland, OR	660,000
4. Population of Oregon	4.3 million
5. Population of the USA	332 million
6. Population of the World	8 billion
7. Number of ants in the world	50 trillion

Find the order of magnitude for each quantity and and plot on ticker below.



4. The volume of a cylinder is given by $V = \pi r^2 h$



- A. If the height were 4 cm and the radius was 3 cm, the volume would be _____.
- B. Write a proportionality relation that relates r and h
- C. Rewrite the volume equation in relative form.
- D. If the height increased by a factor of 1.5 times and the radius changed to only be 75% as much, the volume would change by a factor of _____